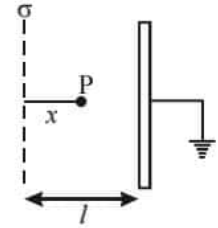


ELECTROSTATICS

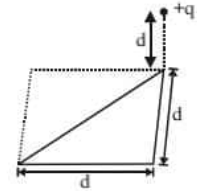
1. A large metallic plate is facing a charged sheet having charge density σ placed parallel to plate at a distance l from the plate. Potential at point P at a distance x from the sheet is

(A) $\frac{\sigma}{2\epsilon_0}(l+x)$ (B) $\frac{\sigma}{2\epsilon_0}(l-x)$
 (C) $\frac{\sigma}{\epsilon_0}(l-x)$ (D) $\frac{\sigma}{\epsilon_0}(l+x)$



2. An isosceles right angle triangle of side d is placed in a horizontal plane. A point charge q is placed at a distance d vertically above from one of the corner as shown in the figure. Flux of electric field passing through the triangle is

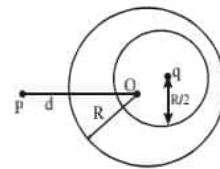
(A) $q/6\epsilon_0$ (B) $q/18\epsilon_0$
 (C) $q/24\epsilon_0$ (D) $q/48\epsilon_0$



3. Potential within a hypothetical charged sphere varies with the distance of a point from centre as $V = -\vec{a} \cdot \vec{r}$ where $\vec{a}$ is a vector of constant magnitude parallel to $\vec{r}$ and $\vec{r}$ is position vector of point under consideration taking centre of sphere as origin. Then the total charge stored within a sphere of radius R is.

(A) $\pi R^2 \epsilon_0 a$ (B) $2\pi R^2 \epsilon_0 a$
 (C) $3\pi R^2 \epsilon_0 a$ (D) $4\pi R^2 \epsilon_0 a$

4. From A conducting sphere of radius R a cavity of radius $R/2$ is cut out as shown in the figure. At the centre of cavity a point charge q is placed then the field and potential at point P as shown in the figure is



(A) $V = \frac{q}{2\pi\epsilon_0 d}, E = \frac{q}{2\pi\epsilon_0 d^2}$ (B) $V = \frac{q}{4\pi\epsilon_0 d}, E = \frac{q}{2\pi\epsilon_0 d^2}$
 (C) $V = \frac{q}{4\pi\epsilon_0 d}, E = \frac{q}{4\pi\epsilon_0 d^2}$ (D) $V = \frac{q}{2\pi\epsilon_0 d}, E = \frac{q}{4\pi\epsilon_0 d^2}$

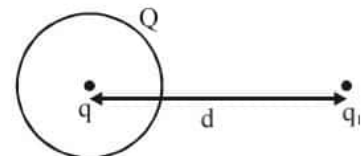
5. A charged particle of mass m and charge q is released from rest from position $(x_0, 0)$ in a uniform electric field $E_0 \hat{j}$. The angular momentum of the particle about origin

(A) is zero (B) is constant
 (C) increases with time (D) decreases with time

6. F_1 is the magnitude of the force on point charge q placed in front of a infinite plane sheet of surface charge density σ . Where F_2 is magnitude of the force on another point charge $-q$ placed in front of infinite conducting plane of surface charge density $-\sigma$. Then F_1 / F_2 is

(A) 1 (B) $\frac{1}{2}$ (C) 2 (D) none of these

7. A thin metallic spherical shell contains a charge Q on it. A point charge q is placed at the centre of the shell and another charge q_1 is placed outside it as shown in the figure. The three charges are positive. Then choose the incorrect option

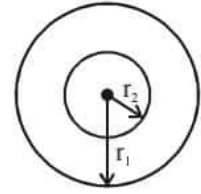


(A) the net force on q is zero
 (B) the force on q due to q_1 is $\frac{1}{4\pi\epsilon_0} \frac{qq_1}{d^2}$ towards left
 (C) the force on q due to Q is non zero
 (D) the force on q due to shell is zero

8. A charge is uniformly distributed inside a spherical body of radius $r_1 = 2r_0$ having a concentric cavity of radius $r_2 = r_0$. (ρ is charge density inside the sphere).

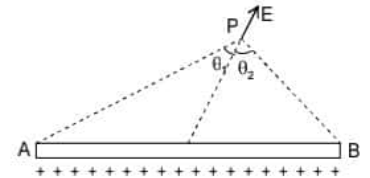
Potential of a point P at a distance $\frac{3r_0}{2}$ from centre is

- (A) $\frac{7\rho r_0^2}{6\epsilon_0}$ (B) $\frac{101\rho r_0^2}{72\epsilon_0}$
 (C) $\frac{17\rho r_0^2}{72\epsilon_0}$ (D) $\frac{84\rho r_0^2}{72\epsilon_0}$

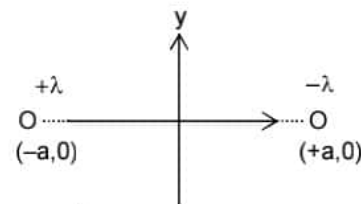


9. The net electric field $\vec{E}$ due to the uniformly charged rod at P makes angles θ_1 and θ_2 with AP and BP respectively. Then θ_1 / θ_2 :

- (A) = 1 (B) > 1
 (C) < 1 (D) $\frac{AP}{PB}$



10. Which E_x - x graph is appropriate for two parallel long straight line charges of linear charge densities $+\lambda$ and $-\lambda$?



- (A) (B)
 (C) (D)

11. Which one of the following does not represent the electrostatic electric field

- (A) $2x\hat{i} + 2y\hat{j} - 2z\hat{k}$ (B) $2x\hat{i} - 2y\hat{j} + 2z\hat{k}$
 (C) $-2x\hat{i} - 2y\hat{j} - 2z\hat{k}$ (D) $2y\hat{i} - 0\hat{j} - 0\hat{k}$

12. A particle of mass m and charge q moves at high speed along the x -axis. It is initially near $x = -\infty$ and it ends up near $x = +\infty$. A second charge Q is fixed at the point $x = 0, y = -d$. As the moving charge passes the stationary charge, its x component of velocity does not change appreciably, but it acquires a small velocity in y -direction. Determine the angle through which the moving charge is deflected

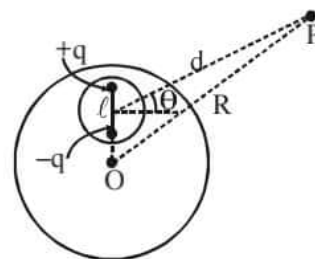
- (A) $\theta = \tan^{-1}\left(\frac{qQ}{2\pi\epsilon_0 d m v^2}\right)$ (B) $\theta = \sin^{-1}\left(\frac{qQ}{2\pi\epsilon_0 d m v^2}\right)$
 (C) $\theta = \tan^{-1}\left(\frac{qQ}{4\pi\epsilon_0 d m v^2}\right)$ (D) $\theta = \sin^{-1}\left(\frac{qQ}{4\pi\epsilon_0 d m v^2}\right)$

13. Work done by an external agent to move slowly a charge Q from rim of a uniformly charged horizontal disc of radius a and charge per unit area σ , to center of this disc is

- (A) $\frac{\sigma a Q}{\epsilon_0} \left(\frac{2}{\pi} - \frac{1}{2}\right)$ (B) $\frac{\sigma a Q}{\epsilon_0} \left(\frac{1}{2} - \frac{1}{\pi}\right)$ (C) $\frac{\sigma a Q}{\epsilon_0} \left(\frac{1}{\pi} - \frac{1}{2}\right)$ (D) $\frac{\sigma a Q}{\epsilon_0} \left(\frac{1}{2} - \frac{2}{\pi}\right)$

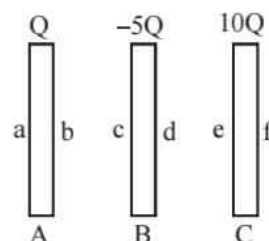
14. A spherical cavity is created in a neutral solid conducting sphere. Inside the cavity, a dipole is placed as shown in the figure. Electrostatic potential at point P only due to charge induced on the inner surface of the cavity is (assume that $\ell \ll d$)

(A) $\frac{q\ell}{4\pi\epsilon_0 d^2} \sin\theta$ (B) $\frac{-q\ell \sin\theta}{4\pi\epsilon_0 d^2}$
 (C) $-\frac{q\ell}{4\pi\epsilon_0 d} \sin\theta$ (D) $\frac{q\ell \sin\theta}{4\pi\epsilon_0 d}$



15. Three very large plates are given charges as shown in the figure. If the cross-sectional area of each plate is the same, then the final charge distribution on the surface of the plates a, b, c, d, e, f is

- (A) 7Q on surface e and 3Q on surface f
 (B) -2Q on surface b and 3Q on surface a
 (C) -7Q on surface d and 2Q on surface c
 (D) the magnitude of charges at surfaces b, c, d, e is equal

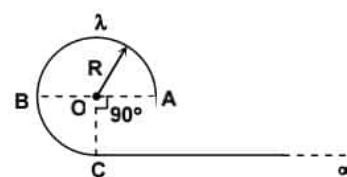


16. The potential energy of a uniform circular disc of radius R and carrying a charge density σ is

(A) $\frac{2}{3} \frac{\pi\sigma^2 R^3}{\epsilon_0}$ (B) $\frac{1}{3} \frac{\pi\sigma^2 R^3}{\epsilon_0}$ (C) $\frac{\pi\sigma^2 R^3}{\epsilon_0}$ (D) can not be calculated

17. A semi infinite line charge of linear charge density λ has the shape as shown in the figure. Portion ABC forms three-fourth of a circle of radius R while the straight portion from C to infinity is parallel to BOA. The field at the centre of circle (O) is

(A) $\frac{\lambda}{2\sqrt{2}\pi\epsilon_0 R}$ (B) $\frac{\lambda}{2\pi\epsilon_0 R}$
 (C) zero (D) None of these

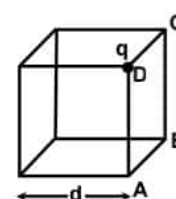


18. Two identical thin rings, each of radius R metres are placed coaxially at a distance R metres apart. If Q_1 coulomb and Q_2 coulomb are respectively the charges uniformly spread on the two rings, then the work done in moving a charge q from the centre of one ring to that of other is

(A) zero (B) $q(Q_1 - Q_2) (\sqrt{2} - 1) / (\sqrt{2} \cdot 4\pi\epsilon_0 R)$
 (C) $q \sqrt{2}(Q_1 + Q_2) / 4\pi\epsilon_0 R$ (D) $q (Q_1 + Q_2) (\sqrt{2} + 1) / (\sqrt{2} \cdot 4\pi\epsilon_0 R)$

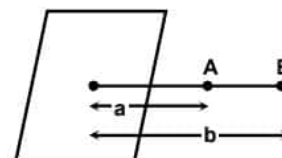
19. A point charge q is placed at corner D of a cube of side d. Flux of electric field passing through face ABCD is

(A) zero (B) $\frac{q}{6\epsilon_0}$ (C) $\frac{q}{12\epsilon_0}$ (D) $\frac{q}{24\epsilon_0}$



20. Two points A and B are at a distance a and b respectively from an infinite metal plate having charge density as σ C/m². Potential difference between A and B is

(A) $\frac{\sigma}{\epsilon_0} (b - a)$ (B) $\frac{\sigma}{\epsilon_0} (a - b)$ (C) $\frac{\sigma}{2\epsilon_0} (b - a)$ (D) $\frac{\sigma}{2\epsilon_0} (a - b)$



21. Two identical conducting sphere having unequal positive charges q_1 and q_2 are separated by distance r. If they are made to touch each other and then separated again to the same distance, the electrostatic force between them in this case will be

- (A) less than before (B) same as before (C) more than before (D) zero

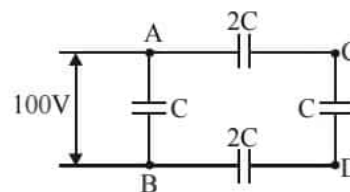
22. Uniform electric field lines pass inward through the surface of a hemisphere of radius R making angle of 30° with the normal to the plane of the surface. The strength of the electric field is E . There is no charge anywhere. The net outward flux through the curved surface is given by
 (A) $\phi_E = E\pi R^2 \cos 30^\circ$ (B) $\phi_E = E\pi R^2 \sin 60^\circ$ (C) $\phi_E = 0$ (D) $\phi_E = 2\pi R^2 E \cos 60^\circ$
23. A particle with charge q and mass m is projected with a velocity v_0 from a large distance towards another identical charged particle at rest. Which of the following is correct?
 (A) The projected charge will move in a circular path about the charge at rest.
 (B) There will be a collision between them.
 (C) The projected particle may come as close as $x = \frac{q^2}{4\pi\epsilon_0 \left(\frac{1}{2}mv_0^2\right)}$ to the particle at rest.
 (D) None of these
24. Three identical particles, each possessing the mass m and charge $+q$, are placed at the corners of an equilateral triangle with side r_0 . The particles are simultaneously set free and start flying apart symmetrically due to Coulomb's repulsion forces. The work performed by Coulomb's forces acting on each particle until the particles fly from one another to a very large distance is.
 (A) $\frac{3kq^2}{r_0}$ (B) $\frac{kq^2}{r_0}$ (C) $\frac{3kq^2}{2r_0}$ (D) $\frac{kq^2}{2r_0}$
 where $k = 1/4\pi\epsilon_0$
25. N capacitors, each of capacitance C are connected in series across a battery of emf ϵ . The charge on each capacitor is Q . Now a dielectric of dielectric constant k is inserted in M capacitors keeping the battery connected ($M < N$) and the new charge happens to be Q' , then
 (A) $\frac{Q}{Q'} = \frac{kM}{(N-M)}$ (B) $\frac{Q'}{Q} = \frac{kN}{k(N-M) + M}$
 (C) $\frac{Q'}{Q} = \frac{kN}{(N+M)}$ (D) $\frac{Q'}{Q} = \frac{kM}{k(N-M) + M}$
26. Following operations can be performed on a capacitor.
 A : connect the capacitor to a battery of emf E .
 B : Disconnect the battery.
 C : Reconnect the battery with polarity reversed.
 D : Insert a dielectric slab into the capacitor.
 Now choose the incorrect option (s)
 (A) in action ABC (perform A, then B and then C), the stored electric energy remains unchanged and no thermal energy is developed
 (B) the charge appearing on capacitor is greater after the action ADB than after the action ABD
 (C) the electric energy stored in the capacitor is greater after the action DAB than the action ABD
 (D) the electric field in the capacitor after the action AD is same as that after the action DA

Multiple Correct

27. A parallel plate capacitor with plate area A and separation d has charge Q . A slab of dielectric constant k is inserted in space between the plates almost completely fills the space. If E_0 and C_0 be the electric field and capacitance before inserting the slab, then
 (A) the electric field after inserting the slab is $\frac{E_0}{k}$
 (B) the capacitance after inserting the slab is $k C_0$
 (C) the induced charge on the slab is $Q\left(1 - \frac{1}{k}\right)$
 (D) the energy stored in the capacitor becomes $\frac{U_0}{k}$, U_0 being the energy of the capacitor before inserting the slab

28. In the circuit shown, a potential difference of 100 V is applied across AB. Then

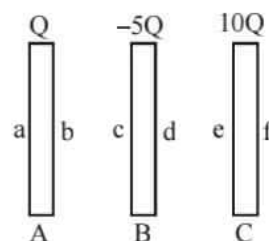
(A) the potential difference between points C and D is 50 V
 (B) the potential difference between A and C is 25 V
 (C) the potential difference between B and D is 25 V



(D) all the capacitors have equal charge

29. Three very large plates are given charges as shown in the figure. If the cross-sectional area of each plate is the same, then the final charge distribution on the surface of the plates a, b, c, d, e, f is

(A) 7Q on surface e and 3Q on surface f
 (B) -2Q on surface b and 3Q on surface a
 (C) -7Q on surface d and 2Q on surface c



(D) the magnitude of charges at surfaces b, c, d, e is equal

30. An electric kettle heater has two coils when one coil is switched on, the water in the kettle begins to boil after 15 minutes, when the other coil is switched on the water boils in 30 minutes.

(A) The water will boil in 45 minutes if the coils are connected in series.
 (B) The water will boil in 10 minutes if the coils are connected in parallel.
 (C) Water boils in 22.5 minutes if coils are connected in series.
 (D) Water boils in 20 minutes if coils are connected in parallel.

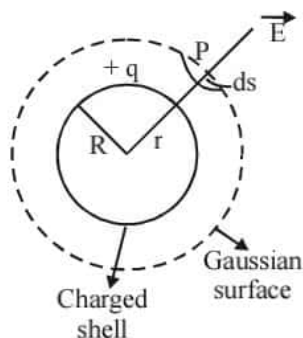
Comprehension

Comprehension-1

Gauss's theorem states that the total electric flux through a closed surface enclosing a charge is equal to $1/\epsilon_0$ times the magnitude of the charge enclosed (ϵ_0 is the permittivity of free space)

$$\phi = \int_s \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

Electric field due to a line charge : - If E is the magnitude of electric field, electric flux through the gaussian surface is given by



$$\phi = \int_s \vec{E} \cdot d\vec{s} = E \cdot 2\pi r l = q / \epsilon_0$$

By choosing proper gaussian surface we can find electric field due to different charge systems. Normally the shape of gaussian surface is shape of charge distribution.

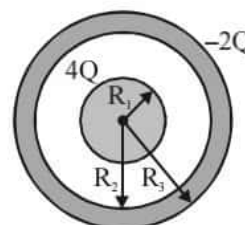
31. For a uniformly charged hollow metallic sphere, the electric field $\vec{E}$, and the electric potential V
- (A) are zero outside it
 (B) vary as $1/r^2$ and $1/r$ respectively at points outside the sphere
 (C) vary as $1/r^2$ and $1/r$ respectively at points inside the sphere
 (D) are non zero inside it

32. A non-conducting solid sphere of radius R is uniformly charged. The magnitude of the electric field due to the sphere at a distance r from its centre.
 (A) increases as r increases, for $r > R$
 (B) decreases as r increases, for $0 < r < \infty$
 (C) decreases as r increases, for $R < r < \infty$
 (D) is discontinuous at $r = R$
33. The electric strength of air is 3.0×10^6 N/C. The largest charge that a 0.30 cm radius metal sphere can hold without sparking is
 (A) 3 nC (B) 8.2 nC
 (C) 9 nC (D) 6 nC
34. A charge Q is distributed over two concentric hollow spheres of radii r and R ($r < R$) such that the surface densities are equal. The potential at the common center is
 (A) $\frac{Q(R^2 + r^2)}{4\pi\epsilon_0(R + r)}$ (B) $\frac{Q}{R + r}$
 (C) Zero (D) $\frac{Q(R + r)}{4\pi\epsilon_0(R^2 + r^2)}$

Comprehension-2

Gauss's law is very useful in finding electric field strength when the charge distribution has a high degree of symmetry. In particular, it is very convenient to apply Gauss's law so as to obtain electric field strength at various points if the charge distribution possesses spherical or cylindrical symmetry. Gauss's law states that electric flux through a closed surface is

equal to $\frac{q_m}{\epsilon_0}$, where q_m is the net charge inside the Gaussian surface.



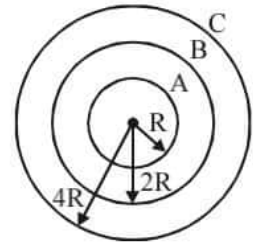
Here we consider a system that has a spherically symmetric charge distribution so that Gauss's law could be conveniently applied. Consider a solid insulating sphere of radius R_1 . It carries a positive charge $4Q$ that is uniformly distributed throughout its volume. A conducting spherical shell of inner radius R_2 , outer radius R_3 , and carrying a charge $-2Q$, is placed concentric with the solid insulating sphere, as shown in figure. (Let r = distance of point from centre)

Answer the following questions :

35. Electric field strength at $r > R_3$ can be expressed as $E =$
 (A) $\frac{1}{4\pi\epsilon_0} \frac{4Q}{r^2}$ (B) $\frac{1}{4\pi\epsilon_0} \frac{3Q}{r^2}$
 (C) $\frac{1}{4\pi\epsilon_0} \frac{2Q}{r^2}$ (D) $\frac{1}{4\pi\epsilon_0} \frac{4Q}{r_3^2}$
36. Electric field strength at r , such that $r < R_1$, can be expressed as $E =$
 (A) $\frac{1}{4\pi\epsilon_0} \frac{4Q}{r^2}$ (B) $\frac{1}{4\pi\epsilon_0} \frac{4Qr}{R_1^3}$
 (C) $\frac{1}{4\pi\epsilon_0} \frac{4Q}{R_1^2}$ (D) zero
37. Charge on inner surface of the conducting spherical shell is
 (A) zero (B) $-4Q$
 (C) $+2Q$ (D) $-5Q$

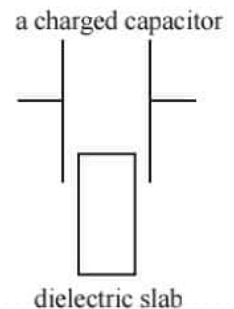
Comprehension-3

Three concentric conducting shells are arranged as shown. Shell B is given charge q_0 . Shell A and C are initially uncharged.



38. If A and C are connected to earth, then charge flow into A and C from earth respectively is
- (A) $\frac{-q_0}{3}, \frac{2q_0}{3}$ (B) $\frac{-q_0}{3}, \frac{-2q_0}{3}$
- (C) $\frac{-2q_0}{3}, \frac{-2q_0}{3}$ (D) $\frac{-q_0}{3}, \frac{-4q_0}{3}$
39. Potential of B in previous case will be
- (A) $\frac{3q_0}{24\pi\epsilon_0 R}$ (B) $\frac{3q_0}{48\pi\epsilon_0 R}$
- (C) $\frac{q_0}{24\pi\epsilon_0 R}$ (D) $\frac{3q_0}{16\pi\epsilon_0 R}$
40. If A and C are disconnected with earth and now B is connected to earth, then charge on B will be
- (A) $\frac{2q_0}{3}$ (B) q_0
- (C) $\frac{q_0}{3}$ (D) $-q_0$

41. A parallel plate air capacitor is charged by connecting its plates to the terminals of a battery. The battery is disconnected and a dielectric slab is introduced partially between the plates, as shown in the figure. Consider the change in the value of each quantity mentioned in the first column below from the time when no dielectric slab was introduced to the time when it was, and match it with the nature of change in it as mentioned in column on the right.



Column A		Column B	
(A)	Surface density of charge, σ	(p)	Increases.
(B)	Electric field intensity, E	(q)	Decreases.
(C)	Charge on the capacitor, q	(r)	Remains same.
(D)	Net force acting on either plate, F	(s)	Increases at some points decreases at others.

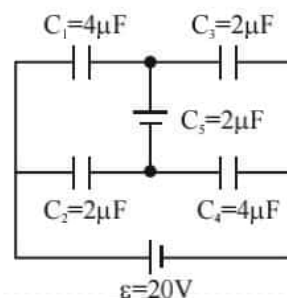
42. A parallel plate capacitor of capacity C_0 is charged by a battery. After charging battery is disconnected a separation between the plates is reduced and a dielectric is filled between the plates. Column I represents a quantity and Column II represent the effect of changes made to capacitor

Column I		Column II	
(A)	P.E. stored by capacitor	(p)	Increase
(B)	Potential differences between plates	(q)	Decrease
(C)	Capacity of capacitor	(r)	Remains same
(D)	Charge on capacitor	(s)	May increase or decrease

43. Four identical conducting plates are kept parallel to each other. Area of each plate is A and d is separation between successive plates. Column I represent arrangement of plates. Column II represent equivalent capacity of arrangement between 1 and 2.

	Column I		Column II
(A)		(p)	$\frac{\epsilon_0 A}{2d}$
(B)		(q)	$\frac{3\epsilon_0 A}{d}$
(C)		(r)	$\frac{3}{2} \frac{\epsilon_0 A}{d}$
(D)		(s)	$\frac{2\epsilon_0 A}{d}$

44. Figure shows an arrangement of capacitor and battery



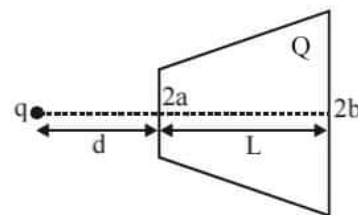
	Column I		Column II
(A)	Charge on C_1	(p)	$24 \mu\text{C}$
(B)	Charge on C_3	(q)	$32 \mu\text{C}$
(C)	Charge on C_5	(r)	$8 \mu\text{C}$
(D)	Charge on C_2	(s)	$16 \mu\text{C}$

45. Match the Following

	Column I		Column II
(A)	If separation between two charge particle is increased P.E. between the charge	(p)	Increases
(B)	If separation between two charge particle is decreases force exerted by one particle on other particle	(q)	Decreases
(C)	On changing separation between the charges, number of electric field lines from the charge	(r)	Remains unchanged
(D)	If the separation between two unequal charge particles decreases. Then potential of a point which lies on a line which is perpendicular bisector to the line joining charges	(s)	May increase or decrease

46. A point charge q is situated at a distance d from one end of a thin non-conducting sheet of constant thickness t and of length L as shown in the figure. The width of the sheet varies linearly from $2a$ to $2b$. The sheet has a charge Q , uniformly distributed. Find the force between the two (in Newton).

Given $d = 1 \text{ cm}$, $a = 18 \text{ m}$, $t = 0.01 \text{ mm}$



$$L = a + b = 144 \text{ m}$$

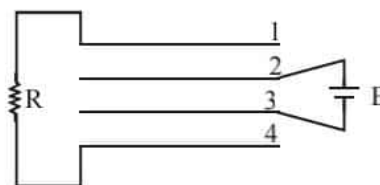
$$Q = 16 \times 10^{-4} \text{ C}, q = 10^{-5} \text{ C}$$

(Do approximation if required)

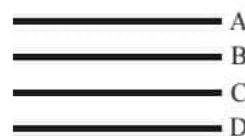
47. Four metallic plates, each having a surface area of one side A , are placed at a distance D from each other. The plates are connected as shown in the figure. Find the ratio of the field strength between plates 2 and 3 and that between plates 1 and 2.

$$\text{Given } R = 3\Omega, E = 72\text{V},$$

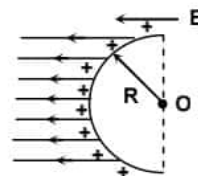
$$D = 0.1 \text{ mm}, A = 4 \text{ m}^2$$



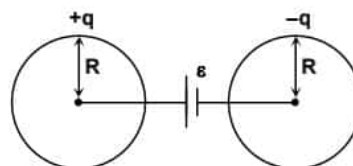
48. Charge Q , $2Q$, $3Q$, $4Q$ are given to four large parallel conducting plates A, B, C, D respectively. B and C are connected to each other by a metallic wire. If the charge that flows from C to B through this wire is nQ . Then find the value of n . The area of each plate is A and separation between plates is d .



49. A semi-circular ring of mass m and radius R with linear charge density λ , hinged at its centre is placed in a uniform electric field as shown in the figure. If the ring is slightly rotated about O and released find the time period (in sec) of oscillation. Take $M = 8 \text{ kg}$, $\lambda = 2\text{C/m}$ and $E = 2\text{N/C}$. Assume that coil is rotated in its own plane.



50. Consider two metallic spheres each of radius R separated by a large distance and connected with a battery of emf ε as shown in the figure. In the electrical equilibrium the charges on the spheres are $+q$ and $-q$. If $q = k(2\pi\varepsilon_0 R\varepsilon)$, then the value of k is



Answers

1. (C)

$$1. \quad E = \frac{\sigma}{\epsilon_0} \therefore V = \frac{\sigma}{\epsilon_0} (l - x)$$

2. (D)

$$\text{flux} = \frac{1}{2} \left(\frac{q}{24\epsilon_0} \right)$$

3. (D)

$$E = -\frac{dV}{dr} = a ; \text{ So flux} = 4\pi R^2 a$$

$$\text{Now } \frac{q_{in}}{\epsilon_0} = 4\pi R^2 a \Rightarrow q_{in} = 4\pi R^2 \epsilon_0 a$$

4. (C)

$$V = \frac{q}{4\pi\epsilon_0 d} ; E = \frac{q}{4\pi\epsilon_0 d^2}$$

5. (C)

$$\vec{v} = \left(\frac{qE}{m} \right) \hat{tj}$$

$$\vec{r} = x_0 \hat{i}$$

$$\vec{L} = m(\vec{r} \times \vec{v})$$

6. (B)

$$F_1 = \frac{\sigma q}{2\epsilon_0}$$

$$F_2 = \frac{\sigma q}{\epsilon_0}$$

$$F_1 / F_2 = \frac{1}{2}$$

7. (D)

8. (B)

$$V_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{\rho \frac{4}{3}\pi(r_2^3 - r_1^3)}{(2r_0)}$$

$$V_{\text{surface}} = \frac{7\rho r_0^2}{6\epsilon_0}$$

From Gauss theorem

$$\int_{V_s}^V dV = - \int_{2r_0}^{3r_0/2} \frac{1}{4\pi\epsilon_0} \frac{q_{\text{enclose}}}{r^2} dr$$

$$V - V_s = - \frac{1}{4\pi\epsilon_0} \int_{2r_0}^{3r_0/2} \rho \frac{4}{3} \pi (r^3 - r_0^3) \frac{1}{r^2} dr$$

Solving

$$V = \frac{101 \rho r_0^2}{72 \epsilon_0}$$

9. A

10. C

11. D

12. A

$$v_x = v = \text{constant}$$

$$q \int E \cos \theta dt = mv_y$$

$$\text{or } q \int E \cos \theta \frac{dA}{2\pi dv} = mv_y$$

$$v_y = \frac{q}{2\pi m dv} \int E \cos \theta dA$$

from Gauss' law

$$\int E \cos \theta dA = \frac{q}{\epsilon_0} \quad \text{Integration for the whole cylindrical surface of radius } d$$

and extending from $x \rightarrow -\infty$ to $x \rightarrow +\infty$

$$\text{Deflection } \theta = \tan^{-1} \left(\frac{qQ}{2\pi \epsilon_0 d m v^2} \right)$$

13. (B)

$$V_R = \frac{\sigma a}{\pi \epsilon_0}$$

$$V_C = \frac{\sigma a}{2\epsilon_0}$$

14. (B)

15. (A), (B), (C)

15. Using the conservation of charge principle, distribute the charge on all the surfaces of the plates a, b, c, d, e, and f. Then use the concept, electric field is zero inside the plates.

16. B

$$\epsilon = \frac{1}{2} \int_0^R \left(\frac{\sigma r}{\epsilon_0} \right) \{ \sigma (2\pi r dr) \}$$

$$= \frac{1}{2} \left(\frac{2\sigma^2 \pi}{\epsilon_0} \right) \int_0^R (r^2 dr) = \frac{\pi \sigma^2 R^3}{3\epsilon_0}$$

17. C

$$dE_{\text{curved}} = \frac{\lambda R d\theta}{4\pi \epsilon_0 R^2}$$

$$dE_{\text{straight}} = \frac{\lambda dx}{4\pi \epsilon_0 (R \sec \theta)^2}$$

$$x = R \tan \theta \Rightarrow dx = R \sec^2 \theta d\theta$$

$$\Rightarrow dE_{\text{curved}} = dE_{\text{straight}}$$

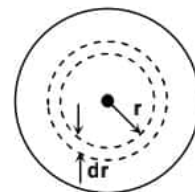
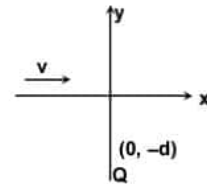
18. B

19. A

Field is perpendicular to area vector so flux is zero.

20. A

$$E = \frac{\sigma}{\epsilon_0}$$



$$\Delta V = \frac{\sigma}{\epsilon_0}(b-a) \Rightarrow V_A - V_B = \frac{\sigma}{\epsilon_0}(b-a) \quad 21. \quad C$$

22.

A

Apply Gauss' law. The net flux is zero but outward flux is given by (A)

23.

C

24.

B

Since the given system is closed, the increment in KE is equal to decrease in P.E.

$$\Rightarrow \frac{3}{2}mv^2 = \frac{2kq^2}{r_0} - \frac{3kq^2}{r}$$

$$\Rightarrow v = \sqrt{\frac{2kq^2(r-r_0)}{mrr_0}}, \quad v \text{ will be max when } r \rightarrow \infty$$

$$\Rightarrow v_{\max} = \sqrt{\frac{2kq^2}{mr_0}}$$

The work performed by the interaction force during the variation of the system's configuration is equal to the decrease in the potential energy

$$W = U_1 - U_2 = \frac{3kq^2}{r_0}$$

$$\therefore \text{Work done per particle} = \frac{kq^2}{r_0}$$

25.

B

26.

(A)

27.

(A), (B), (C), (D)

28.

(A), (B), (C)

29.

(A), (B), (C)

30.

A, B

If r_1 and r_2 are the resistances of the first and the second coils and v = mains voltage.

$$Q = \frac{V^2 t_1}{r_1} = \frac{V^2 t_2}{r_2} \Rightarrow \frac{r_1}{r_2} = \frac{t_1}{t_2} = \frac{1}{2}$$

In series $r_{eq} = r_1 + r_2$

$$\Rightarrow Q = \frac{V^2 t_3}{r_1 + r_2} = \frac{V^2 t_1}{r_1} \Rightarrow t_3 = 3t = 45 \text{ min}$$

In parallel

$$Q = \frac{V^2 t_4 (r_1 + r_2)}{r_1 r_2} = \frac{V^2 t_1}{r_1} \Rightarrow$$

$$t_4 = \frac{r^2}{r_1 + r_2} t = 10 \text{ minutes.}$$

31.

(B)

32.

(C)

33.

(A)

34.

(D)

35.

(C)

36.

(B)

37.

(B)

38.

(B)

39.

(C)

40.

(A)

41.

(A) - (s)

(B) - (q)

(C) - (r)

(D) - (q)

42.

(A) - (q)

(B) - (q)

- (C) – (p)
(D) – (r)

43. (A) – (q)
(B) – (s)
(C) – (r)
(D) – (p)

44. (A) – (q)
(B) – (p)
(C) – (r)
(D) – (p)

45. (A) – (s)
(B) – (p)
(C) – (r)
(D) – (s)

46.
$$F = \frac{1}{4\pi\epsilon_0} \cdot q \cdot \int_{x=d}^{d+L} \frac{Q}{L(a+b)} 2y dx \quad \dots(i)$$

$$y = \frac{b-a}{L}x + a$$

Solving, we get

$$F = \frac{1}{4\pi\epsilon_0} \frac{2Qq}{L(a+b)} \cdot \left[\left(\frac{b-a}{L} \right) \cdot \ln \left(1 + \frac{d}{L} \right) + \frac{aL}{d(d+L)} \right]$$

47. Using conservation of charge principle and induction, do charge distribution on each plate.

48. A negative charge, $-2Q$ flows from B to C.

Solution :

Use conservation of charge principle to find charge on each plates.

49. 6.28

50. 1

$$-\frac{q}{4\pi\epsilon_0 R} + \epsilon = \frac{q}{4\pi\epsilon_0 R}$$

$$\Rightarrow q = (2\pi\epsilon_0 R)\epsilon$$